

SCOPE OF WORK

Project #5 – Water Quality Evaluation of McClusky Canal

Client: Garrison Diversion Project & Red River Valley Water Supply Project

Due Date: May 8, 2026

1. Project Background

The project evaluates water quality from Lake Sakakawea through the McClusky Canal to the Sheyenne River, ultimately affecting Lake Ashtabula. Droughts have reduced water supply in the Red River Valley, so ensuring safe water quality is critical.

2. Objectives

- Assess whether water quality degrades along the canal.
 - Identify the most concerning parameters.
 - Calculate mass loadings for major constituents.
 - Estimate TDS in Lake Ashtabula using a dynamic mass balance model.
 - Provide recommendations for water quality improvement.
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3. Scope of Services

Data Preparation

- Clean and organize water quality data from CSV and Excel files.
- Focus on specified USGS canal locations.

Analysis

- Evaluate major cations (Ca, Mg, Na, SAR), major anions (Cl, SO₄, CO₃, HCO₃), and other parameters (TDS, pH, hardness, temperature).
- Compute mean, median, min, and max at each sampling point.
- Generate boxplots and graphs to show trends along the canal.

Mass Balance

- Apply:

$$Q_{pipe}C_{pipe} + Q_{Sheyenne}C_{Sheyenne} = Q_{Ashtabula}C_{Ashtabula}$$

- Develop a Python function for dynamic input of flow rates and concentrations.

Reporting

- Discuss key water quality concerns.
 - Suggest improvement strategies.
 - Document coding choices and potential bias.
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4. Deliverables

1. Cleaned dataset
 2. Statistical summaries
 3. Boxplots and graphs
 4. Mass balance Python model
 5. TDS and mass loading calculations
 6. Technical report and executive summaries
 7. Gantt chart and timesheet
 8. Annotated code document
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5. Assumptions

- Pipe flow fixed at 6 cfs.
- Mixing in Lake Ashtabula is complete and instantaneous.
- Steady-state flow conditions.